



Sri Lanka Institute of Information Technology

B.Sc. Honours Degree in Information Technology

Specialized in Software Engineering

Final Examination
Year 3, Semester I (2025)

SE3030 – Software Architecture

Duration: 2 Hours

June 2025

Instructions to Candidates:

- ◆ This paper is preceded by a **10-minute** reading time. The supervisor will indicate when answering may commence.
- ◆ This paper has **Four** questions.
- ◆ Answer all questions in the booklet given.
- ◆ The total marks for the paper is **100**.
- ◆ This paper contains 05 pages, including the cover page.
- ◆ Electronic devices capable of storing and retrieving text, including calculators and mobile phones are not allowed.

Question 1**(25 marks)**

You are hired to architect a national digital public health initiative called *MediConnect* by the Health Department, which aims to build an integrated digital platform for managing all public health related data such as medical records, managing prescriptions, scheduling appointments, epidemic outbreaks etc.

The system must integrate with various legacy systems in various hospitals, support both web and mobile interfaces, and be scalable to serve millions of users, including patients, healthcare professionals, and government administrators. It should also accommodate the potential for future features such as AI-based diagnosis support and data analytics. The system is to be implemented in phases, with the first phase being released within a year of project commencement.

The Health Department (the client) has requested your input on choosing the most suitable architectural style for the initial system release and the long-term evolution of the platform. Consider the following architectural patterns: Monolithic, Modular Monolithic, Event-Driven, Microkernel and Microservices to answer the questions below.

1. Identify the functional and non-functional requirements of the *MediConnect* system described above. (2 marks)
2. Based on this the identified requirements, select two most critical quality attributes that should guide architectural decisions. Justify your selection. (4 marks)
3. Analyse how each of the given architectural styles (Monolithic, Modular Monolithic, Event-Driven, Microkernel, Microservices) aligns or conflicts with the quality attributes identified in question 2. You may use a table for this. (4 marks)
4. Recommend one architectural pattern as the primary candidate for the initial release of *MediConnect*. Justify your choice by referring to both system requirements and trade-offs. (4 marks)
5. The client is also concerned about the system's ability to evolve over the next decade as the requirements change over time. Do you think the same architectural style should be maintained or should it be transitioned to another pattern to better cater to the quality attributes identified? Justify your position. (5 marks)
6. Propose a high-level architectural diagram (a block diagram is fine) to illustrate how key components would be organized under your recommended architecture for the initial design. Clearly label responsibilities and interactions among components. Identify relevant communication mechanisms as well. Provide a small description explaining your design. (6 marks)

Question 02**(25 marks)**

A company is considering modernizing an existing desktop-based sales record system into a scalable, maintainable, and cloud-based sales management system. The current system faces problems with meeting the current demands placed on it. Further it encounters technical incompatibilities with modern software and hardware infrastructure. The sales staff find it increasingly frustrating to work with the current system. However, the management of the company has only agreed to allow this modernization, provided the project is well justified and has strict deadlines and budget constraints. They also expect support for a third-party Point-of-Sales (POS) system integration and wants to evolve the system incrementally in response to user feedback. You are being consulted as a software architect regarding this situation.

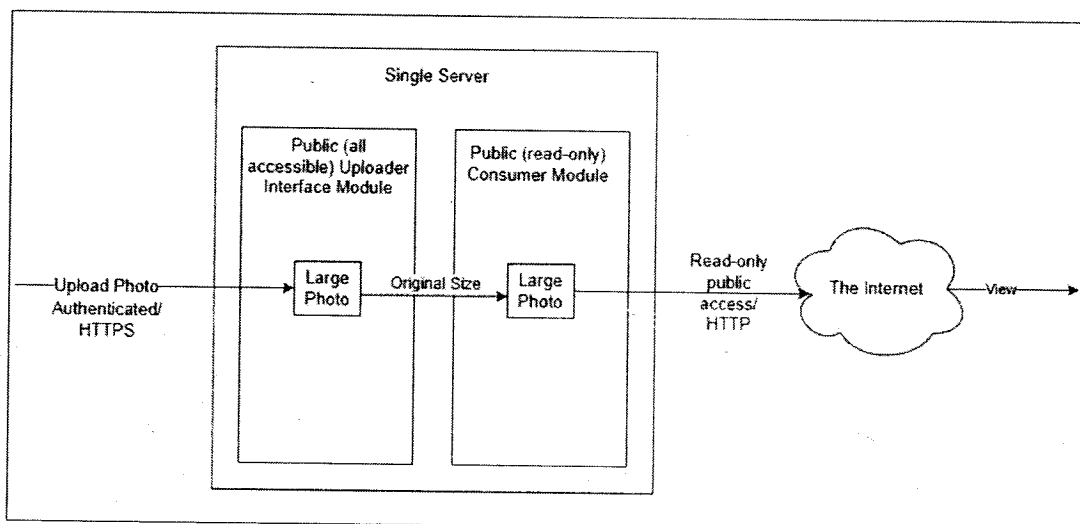
1. Identify three key influences that would shape your architectural decisions in this situation (Hint: consider the *Architecture Business Cycle (ABC)*). (3 marks)
2. Based on *Architectural Activities*, justify the business case for this change in detail with at least five (5) points. (10 marks)
3. Assume that the business case is now approved by the management. Briefly discuss how you would carry out the next three (03) *architectural activities* that should follow in this system. Clearly describe how each activity contributes to progressing the system's modernization. (6 marks)
4. "Ensuring that the implementation conforms to the architecture" is an important step in the *Architectural Activities*. With appropriate reasons, discuss the importance of this step in the *maintenance phase* of the software. (6 marks)

Question 03**(25 marks)**

1. Explain the Architecture Business Cycle. (4 marks)
2. Describe what Non-Functional Requirements are and outline the 3 main categories of Non-Functional Requirements. (4 marks)
3. What is 4+1 View Model and explain the focus areas of that model. (5 marks)
4. Explain Cloud Architecture style and describe the advantages and disadvantages of its different service offerings. (4 marks)
5. Explain SAAM (Software Architecture Analysis Method) in brief and outline its main objectives and benefits. (4 marks)
6. Explain the similarities and differences of N-Tier architecture and Layered architecture. (4 marks)

Question 04**(25 marks)**

- a) A public photo sharing website on a single server needs to be built with the following considerations:
- An Uploader interface
 - For uploading new photos
 - Only a set of identified users can upload content
 - A public consumer interface
 - To view thumbnails of photos
 - To access & download large size photos
 - Expect high availability on accessing large photos
 - Anticipated thumbnail view frequency is 10,000 times more than large photo view frequency
- i) Write concrete Quality Attribute scenarios for Availability for the above requirement. (3 marks)
- ii) Propose a tactic and describe how you can improve Availability for above. (2 marks)
- iii) Write concrete Quality Attribute scenarios for Performance for the above requirement. (3 marks)
- iv) Propose a tactic and describe how you can improve Performance for above. (2 marks)
- v) Critique the given sample architecture design of the photo sharing web site given below in-comparison to the given requirements. Suggest any improvements required with appropriate justifications. (6 marks)



(6 marks)

- b) A courthouse is planning to digitize all its records as a part of a government initiative that will help to back up and safeguard information and for the ease of searching and exchanging digital documents via integrations with different government agencies. In their current document archive, there are millions of documents, and the number of documents keeps increasing as court cases proceed. As part of this larger solution a not-for-profit organization has volunteered to build and operate a digital record management system where they will be scanning and converting the documents into PDF format. Also, they plan to use character and image recognition features assisted with AI to build a smart search in a future initiative, so it'll be easier to search and retrieve the records. For all new records the clients will upload digital records while the existing records are slowly scanned and uploaded over a longer period.
- i) Draw and explain how you would architect the above solution. (6 marks)
 - ii) Identify a Trade-off in above proposal and discuss mitigation plans. (3 marks)

End of the paper